Employee Payroll System - MySQL Capstone Project

Project Objective

Design and implement an Employee Payroll Management System using MySQL to manage employee details, department information, daily attendance and monthly salary payments. The system should calculate employee salaries based on basic salary, allowances and deductions, maintain payroll history and generate useful salary reports.

Problem Statement

An organization wants to digitize its employee payroll operations, including employee information, department details, attendance records and monthly salary payments.

Maintaining these records manually makes it difficult to calculate salaries, track employee attendance, monitor deductions and generate monthly payslips.

Design a relational database that efficiently manages employee information, attendance and payroll processing while maintaining data consistency and generating meaningful reports for the HR and finance departments.

MySQL Workflow

- Requirement Analysis
- Identify Entities
- ER Diagram
- Normalization (1NF → 3NF)
- Create Database
- Create Tables
- Insert Sample Data
- CRUD Operations
- Numeric, String & Date Functions
- GROUP BY & HAVING
- Joins & Subqueries
- Views
- Transactions
- Stored Procedures & Triggers

Database Design

- Departments(dept_id PK, dept_name, location)
- Employees(emp_id PK, emp_name, phone, email, job_title, hire_date, basic_salary, dept_id FK, employment_status)
- Attendance(attendance_id PK, emp_id FK, attendance_date, attendance_status, overtime_hours)

- Payroll(payroll_id PK, emp_id FK, pay_month, basic_salary, allowances, deductions, net_salary, payment_date, payment_status)

Insert Data

INSERT INTO Departments VALUES(1, 'Human Resources', 'Hyderabad'),(2, 'Finance', 'Bengaluru'), (3, 'Information Technology', 'Vijayawada'), (4, 'Marketing', 'Chennai');

INSERT INTO Employees VALUES

(101, 'Aarav Sharma', '9000000101', 'aarav@company.com', 'HR Executive', '2021-06-15', 45000, 1, 'Active'),

(102, 'Bhavya Reddy', '9000000102', 'bhavya@company.com', 'Accountant', '2020-03-10', 52000, 2, 'Active'),

(103, 'Charan Kumar', '9000000103', 'charan@company.com', 'Software Developer', '2022-01-20', 38000, 3, 'Active'),

(104, 'Divya Rao', '9000000104', 'divya@company.com', 'HR Manager', '2019-08-05', 60000, 1, 'Active'),

(105, 'Farhan Ali', '9000000105', 'farhan@company.com', 'Marketing Executive', '2023-04-12', 35000, 4, 'Active'),

(106, 'Geetha Nair', '9000000106', 'geetha@company.com', 'Financial Analyst', '2021-11-18', 42000, 2, 'Active'),

(107, 'Harish Varma', '9000000107', 'harish@company.com', 'System Administrator', '2020-07-22', 48000, 3, 'Active'),

(108, 'Ishita Singh', '9000000108', 'ishita@company.com', 'Marketing Manager', '2018-05-14', 55000, 4, 'Active'),

(109, 'Kiran Teja', '9000000109', 'kiran@company.com', 'HR Assistant', '2024-02-01', 40000, 1, 'Active'),

(110, 'Lavanya Devi', '9000000110', 'lavanya@company.com', 'Junior Accountant', '2023-09-11', 32000, 2, 'Active');

INSERT INTO Attendance VALUES

(501, 101, '2026-10-01', 'Present', 1.5), (502, 101, '2026-10-02', 'Present', 0),(503, 102, '2026-10-01', 'Present', 0), (504, 102, '2026-10-02', 'Absent', 0), (505, 103, '2026-10-01', 'Present', 2), (506, 103, '2026-10-02', 'Present', 0), (507, 104, '2026-10-01', 'Leave', 0), (508, 104, '2026-10-02', 'Present', 1), (509, 105, '2026-10-01', 'Present', 0),(510, 106, '2026-10-01', 'Present', 0.5), (511, 107, '2026-10-01', 'Absent', 0), (512, 108, '2026-10-01', 'Present',

1), (513, 109, '2026-10-01', 'Present', 0), (514, 110, '2026-10-01', 'Present', 0), (515, 110, '2026-10-02', 'Present', 2);

INSERT INTO Payroll VALUES

(601, 101, '2026-10-01', 45000, 5000, 2000, 48000, '2026-10-31', 'Paid'),

(602, 102, '2026-10-01', 52000, 6000, 2500, 55500, '2026-10-31', 'Paid'),

(603, 103, '2026-10-01', 38000, 4000, 1500, 40500, '2026-10-31', 'Paid'),

(604, 104, '2026-10-01', 60000, 8000, 3000, 65000, '2026-10-31', 'Paid'),

(605, 105, '2026-10-01', 35000, 3500, 1000, 37500, '2026-10-31', 'Paid'),

(606, 106, '2026-10-01', 42000, 4500, 1800, 44700, '2026-10-31', 'Paid'),

(607, 107, '2026-10-01', 48000, 5000, 2200, 50800, '2026-10-31', 'Paid'),

(608, 108, '2026-10-01', 55000, 7000, 2800, 59200, '2026-10-31', 'Paid'),

(609, 109, '2026-10-01', 40000, 4000, 1200, 42800, '2026-10-31', 'Pending');

Business Reports

1. Display all employee details from the Employees table.
2. Display employee names, job titles and basic salaries of employees belonging to department ID 2.
3. Retrieve the distinct job titles available in the organization.
4. Find employees whose employment status is either Active or Inactive using the IN operator.
5. Display employees whose basic salary is between ₹35,000 and ₹50,000 using BETWEEN.
6. Find employees who belong to department ID 1 or department ID 3 and whose basic salary is greater than ₹40,000.
7. Calculate the total number of employees working in each department.
8. Find the average, maximum and minimum basic salary for each department.
9. Calculate the total salary expenditure of the organization using the net_salary column in the Payroll table.
10. Display the ceiling and floor values of each employee's net salary.
11. Calculate the percentage contribution of each employee's net salary towards the total monthly payroll expenditure. Round the result to two decimal places.
12. Display employee names in uppercase and their email addresses in lowercase.
13. Generate an employee identification display by concatenating the employee’s name and employee ID. Example: Aarav Sharma - 101.

14. Display the length of each employee's name and extract the first four characters of their name.
15. Display employee names along with their completed years of experience in the organization using an appropriate date function.
16. Display attendance dates with their corresponding month names and day names.
17. Calculate the number of days each employee has been working in the organization from their hire date.
18. Display a combined list of employee names and department names using UNION.
19. Display employee IDs who have both attendance records and payroll records using INTERSECT.
20. Find employee IDs who are present in the Employees table but do not have any payroll records using EXCEPT.
21. Display employee names along with their department names and locations.
22. Display employee names, attendance dates and attendance statuses for all attendance records.
23. Generate a monthly salary report displaying employee name, department name, basic salary, allowances, deductions and net salary.
24. Display employee names, their attendance dates and overtime hours, including employees who do not have any attendance records.
25. Display department names along with the total number of employees and their combined basic salaries, including departments that have no employees.
26. Find employees whose basic salary is greater than the average basic salary of all employees.
27. Find the employee or employees who receive the highest net salary.
28. Display employees whose net salary is greater than the average net salary of their respective departments.
29. Find employees who have not received any payroll records using NOT EXISTS.
30. Find departments whose total monthly net salary expenditure is greater than the average department-wise salary expenditure.
31. Create a view named Employee_Details_View to display employee name, job title, department name, hire date and basic salary.
32. Create a view named Monthly_Payroll_View to display employee name, department name, pay month, basic salary, allowances, deductions and net salary.
33. Demonstrate COMMIT by updating an employee's payroll allowance and corresponding net salary, then permanently save the changes.
34. Demonstrate SAVEPOINT and ROLLBACK by performing two payroll updates within a single transaction. Use a savepoint to reverse the second update while retaining the first update, and then commit the changes.
35. Create a stored procedure that accepts a department ID as an IN parameter and displays all employees belonging to that department along with their job titles and basic salaries.
36. Create a stored procedure that accepts an employee ID as an IN parameter and displays the employee's attendance details, including present days, absent days and leave days.

37. Create a stored procedure that accepts a department ID as an IN parameter and returns the total monthly payroll expenditure of that department using an OUT parameter.
38. Create a BEFORE INSERT trigger on the Payroll table to automatically set allowances to 2000 if the allowance value is NULL.
39. Create a BEFORE INSERT trigger on the Attendance table to automatically set the attendance status to 'Present' if no status is provided.